

A good LDP rate with nonconvex level sets

Disproof of the unqualified convexity assertion in Conjecture 00000002821

Independent formal solution

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Abstract

A constant sequence of symmetric two-point probability measures satisfies a large deviation principle at speed $n + 1$ with rate zero at -1 and 1 , and $+\infty$ elsewhere. This rate is proper, nonnegative, lower semicontinuous, and good. Every finite nonnegative sublevel is the compact but nonconvex set $\{-1, 1\}$; its exact zero level is the same set. Thus the assertion that LDP rate level sets are always compact and convex is false, even after adding goodness. The accompanying Lean project verifies the actual probability measures and the complete open/closed LDP bounds, using the extended logarithm at probability zero. It does not assign invented meanings to the undefined extreme-point counting or affine-degeneration wording.

1 Source and precise logical scope

The supplied English source states:

Definition: Compactness and convexity of LDP level sets: the regularity layer of rate functions. Conjecture: Convex-hull rigidity of level sets: level sets of the rate are compact and convex, with the number of extreme points of the hull an exponential-type finite value of the level; the extreme-point structure degenerates for affine rates. (level set extreme point spectrum)

The supplied Chinese version likewise asserts compactness and convexity of the rate's level sets, then the finite exponential-type count and affine degeneration. Both versions were read before selecting the example. Neither specifies a Cramér setting, sample means, a convexity hypothesis on the rate, an ambient finite-dimensional restriction beyond an unspecified space, a formula for “exponential-type,” or a definition of “degenerates.”

We refute the explicit compact-and-convex assertion under the standard meaning of an LDP rate. An example on the usual real line suffices for an unqualified claim. The necessary consequence being refuted is:

$$\begin{aligned} &\text{For every sequence of probability measures satisfying an LDP} \\ &\text{with a proper good rate } I, \quad \{x : I(x) \leq r\} \text{ is convex for every finite } r \geq 0. \end{aligned} \tag{1}$$

Adding “good” and “proper” here narrows the class of potential counterexamples: the example meets both conditions. Thus it refutes the unqualified assertion as well as this more restricted necessary consequence. We also handle the alternate convention that “level set” means $\{x : I(x) = r\}$, by a nonconvex exact zero level. For sublevels, the obstruction occurs at every finite positive level as well, so it is not confined to the boundary level zero.

The nouns following “Definition” do not provide a mathematical hypothesis restricting the problem to convex rates or already-convex sublevels. If such a restriction was intended, it is an additional hypothesis absent from the supplied text, and our result does not purport to settle that amended problem. Nor do we claim a formalization of the undefined counting or degeneration phrases. No such choice is necessary: a conjunction containing the false convexity assertion is false. We make no independent assertion here about an extreme-point growth law or affine degeneration.

2 Definitions and conventions

The state space is $\mathbb{R}$ with its usual topology and Borel sigma algebra. A rate function takes values in $[0, +\infty]$ and is lower semicontinuous. We call it *proper* when it is finite at some point. We call it *good* when every sublevel $\{I \leq r\}$ for finite $r \geq 0$ is compact. The extended logarithm has $\log 0 = -\infty$ and agrees with the ordinary real logarithm on positive finite arguments. Infima of empty sets of rates are $+\infty$. All limits of logarithmic probabilities below take place in the extended real line $[-\infty, +\infty]$ with its order topology.

For a positive speed $a_n \rightarrow +\infty$, the LDP means that for every open $G \subseteq \mathbb{R}$ and every closed $F \subseteq \mathbb{R}$,

$$-\inf_{x \in G} I(x) \leq \liminf_{n \rightarrow \infty} \frac{1}{a_n} \log \mu_n(G), \quad (2)$$

$$\limsup_{n \rightarrow \infty} \frac{1}{a_n} \log \mu_n(F) \leq -\inf_{x \in F} I(x). \quad (3)$$

This is the definition encoded by **HasLDP**. It additionally records that every μ_n is a probability measure, the speed is positive and tends to infinity, and I is lower semicontinuous. The formal infimum is an infimum in $[0, +\infty]$, followed by its canonical embedding in the extended reals and negation. In particular, empty sets and infinite rates are included.

3 The example and its regularity

Set

$$S = \{-1, 1\}, \quad \mu = \frac{1}{2}\delta_{-1} + \frac{1}{2}\delta_1, \quad \mu_n = \mu \quad (n \in \mathbb{N}), \quad a_n = n + 1,$$

and define

$$I(x) = \begin{cases} 0, & x \in S, \\ +\infty, & x \notin S. \end{cases}$$

Here $\mathbb{N}$ starts at zero. The shift in the speed ensures every denominator is positive, while $a_n \rightarrow +\infty$. Dirac measures are Borel probability measures and the two weights sum to one, hence every μ_n is a genuine probability measure. Equivalently, one may realize the same laws by taking one symmetric random variable with values -1 and 1 and using it at every index; no independence or sample-mean assumption is part of the LDP.

Lemma 1. *The function I is a proper good rate. For every finite $r \geq 0$, $\{I \leq r\} = S$, and $\{I = 0\} = S$.*

Proof. At either point of S , the rate is zero, so I is proper and its minimum is zero. Off S , it is infinite. These observations give the two stated set equalities. The set S is finite, hence compact and closed. All finite nonnegative sublevels are consequently compact and closed; the sublevel at $+\infty$ is the whole space. Thus every sublevel in the ordered codomain $[0, +\infty]$ is closed, which is the

closed-sublevel characterization of lower semicontinuity. This proves goodness and the rate-function property. $\square$

4 Verification of the genuine LDP

Lemma 2. *For a finite $p \in [0, +\infty)$,*

$$\frac{1}{n+1} \log p \longrightarrow \begin{cases} -\infty, & p = 0, \\ 0, & p > 0. \end{cases}$$

Proof. If $p = 0$, the extended logarithm is $-\infty$. Multiplication by the strictly positive finite number $1/(n+1)$ leaves it $-\infty$ at every index. If $p > 0$, its logarithm is a finite real constant. The real sequence $1/(n+1)$ tends to zero, so the real product tends to zero. The canonical embedding of the reals into the extended reals is continuous, giving the stated extended-real limit. In particular, a totalized real logarithm at zero is never used in this argument. $\square$

Proposition 3. *For every set $A \subseteq \mathbb{R}$, including the empty set,*

$$\lim_{n \rightarrow \infty} \frac{1}{n+1} \log \mu_n(A) = - \inf_{x \in A} I(x).$$

For nonmeasurable sets the left side uses the measure's outer-measure extension; the LDP itself uses only open and closed sets.

Proof. For any set A , the two Dirac atoms give

$$\mu(A) = \frac{1}{2} \mathbf{1}_{\{-1 \in A\}} + \frac{1}{2} \mathbf{1}_{\{1 \in A\}}.$$

If $A \cap S \neq \emptyset$, then $\mu(A)$ is $1/2$ or 1 , and $\inf_{x \in A} I(x) = 0$: nonnegativity gives the lower bound and an atom in A attains zero. Lemma 2 gives limit zero, as required. If $A \cap S = \emptyset$, then $\mu(A) = 0$ and every rate in A is infinite, so its infimum is $+\infty$. This remains true for $A = \emptyset$ under the empty-infimum convention. The normalized logarithm is identically $-\infty$ and has that limit, which is precisely minus the rate infimum. $\square$

Theorem 4. *The sequence (μ_n) satisfies the LDP at speed $a_n = n+1$ with the proper good rate I defined above.*

Proof. The probability, speed, and regularity conditions were checked above. For each open G , Proposition 3 identifies its logarithmic limit, and hence its $\liminf$, with $-\inf_G I$; this gives (2) with equality. For each closed F the same proposition identifies its $\limsup$ with $-\inf_F I$, giving (3) with equality. No set or endpoint exception is required. $\square$

5 Failure of convexity

Theorem 5. *Every finite nonnegative sublevel of the proper good LDP rate I above is nonconvex. Its exact zero level is also nonconvex. Consequently (1), and hence the source's unqualified compact-and-convex assertion, is false.*

Proof. Both -1 and 1 belong to S , but their midpoint $\frac{1}{2}(-1) + \frac{1}{2}(1) = 0$ does not. Thus S is not convex. Lemma 1 identifies every indicated sublevel and the exact zero level with S . Theorem 4 verifies all LDP hypotheses. For the formal negation of the universal statement, take the positive finite level $r = 1$ in this example. $\square$

6 Formal verification and reproducibility

The project pins Lean v4.19.0 and Mathlib revision

c44e0c8ee63ca166450922a373c7409c5d26b00b.

All authored mathematical source is contained in these four files:

Conjecture2821/Basic.lean

Defines the actual measure, rate and support; proves probability mass one, properness, lower semicontinuity, compactness of all finite sublevels, the two nonconvexity statements, and the two possible rate infima.

Conjecture2821/Limits.lean

Proves the general normalized extended-logarithm limit in Lemma 2, with the zero-probability case explicit.

Conjecture2821/LDP.lean

Defines the standard LDP bounds, proves the valid speed and the eventwise limit, and assembles the LDP and disproof. The main declarations are `eventwise_limit`, `laws_have_LDP`, `counterexample`, and `not_all_good_ldp_sublevels_convex`.

Conjecture2821.lean

Imports the complete proof and prints the axiom dependencies of the main results and supporting regularity statements.

The theorem `counterexample` has no mathematical hypotheses: it proves, for the displayed definitions, the conjunction of an actual LDP, goodness, properness, nonconvexity of every finite nonnegative sublevel, and nonconvexity of the exact zero level. The final negation theorem quantifies over arbitrary measure sequences, speeds, and nonnegative extended-real rates on $\mathbb{R}$. Its assumptions are the exact LDP predicate, goodness, and properness; its refuted conclusion is convexity of every finite nonnegative sublevel. Thus the formal theorem does not assume any disputed conclusion or replace the rate with an unrelated model.

From this directory, with the pinned toolchain and dependency available, run `lake build`, or run `bash verify.sh` for a full build and a warnings-as-errors replay of every authored Lean file. A first setup may use Mathlib’s normal `lake exe cache get` to obtain compiled dependencies. The committed manifest pins all transitive revisions; no local absolute path is needed by the portable project. The local authoring environment used read-only stock dependencies and a private build overlay for the one missing unchanged stock module, `Mathlib.Analysis.SpecialFunctions.Log.ENNRealLog`.

Successful compilation and a separate warnings-as-errors replay were performed on all authored files. The axiom lists for the logarithmic limit, the LDP, the counterexample, and the universal negation contain exactly `propext`, `Classical.choice`, and `Quot.sound`, the standard logical foundations used by Mathlib. There are no added axioms, admissions, unchecked mathematical roots, or numerical decision shortcuts. No numerical computation or mathematical auxiliary script is needed; the build scripts only orchestrate Lean compilation and preserve logs. Failed main-author attempts are retained in original logs separately from successful verification logs. A helper’s first two failure-log files were overwritten during its initial iterations; its evidence explicitly labels the preserved transcript copies and fresh reproductions, rather than calling them original raw captures. None of these failures is counted as a successful check.

7 Provenance and limitations

The exact bilingual source, the two contribution-rule files, and the pinned configuration were supplied through a clean input bundle. Its checksum manifest has SHA-256

67647a5e3e405c079ce8f515bc45528a29bb3be717027254d86acadf4b5eeb3f.

The exact source file has SHA-256

65fad91a6f07e55a7fb2d2d9919044df04597c6b5ace3d8780cf4abc59edc876.

The input files are frozen separately with their individual checksums. Only that bundle, the allowed stock Lean/Mathlib distribution, and the new current-task authoring files were used. A clean helper derived only the normalization lemma from the same inputs and the current independently developed two-atom construction; its source was inspected and independently recompiled in the main private build. No previous solution, repository history, other conjecture, source review, selector output, or internet material informed this argument. Compiler logs and authored-source hashes accompany the evidence package.

The mathematical limitation is the source's unspecified extra wording: this report does not formalize an exponential counting formula or a meaning of degeneration, and makes no claim about a different conjecture in which convexity is imposed as a hypothesis. The source's explicit unqualified convexity assertion is already disproved by the full LDP above.